

SCIENTIFIC REVIEW AND REVISION PLAN

The LUMINA Project: Probing Hydrogen and Helium Reionization with Line Intensity Mapping

Manuscript version: 17 August 2026

BOTTOM LINE Recommendation: major revision before submission

The project is important and publishable. The current manuscript should not be submitted unchanged because three technical questions could alter central results: the normalized bispectrum definition and implementation, the spatial Ly-alpha background calculation, and the amplitude and physical origin of the 3.46 cm signal. These are localized, testable issues rather than reasons to abandon the paper.

Strongest contribution	A uniquely large RHD volume linking late hydrogen reionization, helium reionization and line-intensity statistics.
Highest-risk results	Bispectrum amplitudes and geometry, spatial spin-coupling fields, quantitative 3He+ amplitudes, Cosmic Dawn timing and detectability claims.
Likely robust after revision	The value of the 500 cMpc volume, the qualitative late-EoR hydrogen morphology, the weak global helium coupling, and the importance of rare quasar environments.
Best publication strategy	Lead with large-volume topology and late-EoR statistics; present Cosmic Dawn and 3He+ amplitudes as model-dependent reference predictions until the validation tests are complete.

Prepared as a constructive pre-submission assessment for manuscript revision

How to use this report

The report separates result-changing checks from interpretive and editorial improvements. The postdoc should work through the priorities in order: P0 items first, then P1, then P2. A polished rewrite should begin only after the P0 validation results are known, because those results determine which figures, claims and conclusions survive unchanged.

- P0 - validity blockers: issues that may require recalculation or figure regeneration.
- P1 - scientific credibility: validation and documentation needed for referees to trust the results.
- P2 - interpretation and presentation: wording, notation and structure that should be corrected before submission.

Contents

- 1. Executive scientific assessment
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WORKFLOW Recommended revision workflow

First verify the bispectrum and Ly-alpha algorithms. Second decompose the 3He^+ signal and repeat it with gas masks. Third revise the physical interpretation and observational language. Only then rewrite the abstract, conclusion and figure captions.

1. Executive scientific assessment

The manuscript contains a potentially high-impact result set. A 500 cMpc radiation-hydrodynamic calculation with galaxy formation, black-hole growth, hydrogen and helium ionization, X-ray sources and line-intensity statistics is a major numerical achievement. The large volume is genuinely valuable for low- k modes, rare quasars and the high-radius tail of ionized structures. The manuscript also benefits from an explicit attempt to compute hydrogen and helium spin temperatures rather than imposing saturation everywhere.

The strongest and most defensible part of the paper is the late hydrogen-reionization analysis, especially where the hydrogen spin temperature is well above the CMB and the brightness field is controlled mainly by neutral fraction, density and velocity structure. The comparison with THESAN-1 and the use of one-, two- and three-point statistics are useful. The broad conclusion that the diffuse 3He^+ spin temperature remains close to the CMB is also physically plausible.

The current difficulty is not a lack of interesting science. It is that several of the most novel quantitative claims rest on algorithms or interpretations that are not yet sufficiently demonstrated. The normalized bispectrum equation appears dimensionally inconsistent; the written Ly-alpha background method may use the wrong radial kernel; the 3He^+ power appears high compared with recent weak-coupling predictions; and the observational-detectability and convergence language is stronger than the evidence supports.

What is likely to survive the revision

- The 500 cMpc volume substantially improves sampling of rare quasar environments and long-wavelength modes.
- Late-EoR hydrogen morphology is highly non-Gaussian and is better characterized by combining the VID, power spectrum, bispectrum and real-space bubble statistics.
- Hydrogen and helium reionization exhibit different source demographics and characteristic spatial scales in the adopted model.

- Diffuse 3He⁺ emission is intrinsically weak because the spin temperature remains close to the CMB over most of the volume.
- The large-volume simulation is a valuable theoretical reference even where detailed microphysics remains model dependent.

What may change after the required checks

- The amplitude and angular dependence of every normalized bispectrum curve.
- The inferred relation between bispectrum zero crossings and bubble radii.
- The spatial topology of hydrogen and helium spin coupling if the Ly-alpha kernel is incorrect or overly smoothed.
- The 3He⁺ VID, power spectrum and bispectrum if dense effective-ISM cells dominate the signal.
- The timing and depth of the Cosmic Dawn signal once X-ray energy partition is treated self-consistently.
- All statements about present-day detectability and numerical convergence.

2. Revision priority matrix

Priority	Issue	Why it matters	Minimum action before submission
P0	Bispectrum normalization and triangle geometry	A code-level error would change all amplitudes and angular dependence; several figure labels are mathematically inconsistent.	Verify the BiFFT normalization used in production, correct Eq. (4), regenerate figures if needed, and validate zero-crossing scales on synthetic fields.
P0	Spatial Ly-alpha kernel and X-ray energy bookkeeping	The written FFT method appears to use a cumulative volume top-hat for a fixed-time shell contribution; secondary excitation may also be double counted with $f_{\text{heat}} = 1$.	Document the actual shell estimator, compare with direct summation, run point-source and uniform-field tests, and explicitly close the X-ray energy budget.
P0	Helium 3.46 cm amplitude and literature comparison	The predicted power appears substantially above recent weak-coupling calculations. This may reflect real source physics or a modelling artifact.	Provide a component decomposition, direct comparison to recent work, one-zone coupling benchmarks, and gas-phase masking tests.
P1	Dense gas and effective-ISM contamination	The signal is concentrated near quasars and dense gas, where the effective two-phase ISM temperature is not a literal gas temperature for atomic rates.	Repeat helium VID, power spectrum and bispectrum after masking star-forming cells, halos and high-overdensity gas.
P1	Detectability language	Crossing an idealized thermal-noise curve is not equivalent to recoverability in real data.	Replace current-detection claims with idealized sensitivity language unless instrument forward modelling is added.
P1	Convergence and cosmic-variance terminology	Subvolumes of one parent box are not independent realizations and do not prove convergence.	Use "subvolume scatter" and "reduced finite-volume effects"; reserve "converged" for explicit resolution/realization tests.
P1	Low-redshift radiation treatment	Below $z = 4.75$ the calculation uses one AGN-dominated bin plus a uniform H I/He I background, limiting spectral hardening and full H-He coupling claims.	Disclose this prominently and add switch-redshift and group-averaging sensitivity tests.

Priority	Issue	Why it matters	Minimum action before submission
P1	RSD, gridding and Fourier reproducibility	VID tails and small-scale higher-order statistics can depend strongly on mapping, clipping, window functions and masks.	Document coordinate mapping, velocity-gradient treatment, mass assignment, deconvolution, line-of-sight averaging, binning and covariance.
P2	Statistical and topological interpretation	Several claims over-map one-point or Fourier statistics onto connectivity, shape and percolation.	Correct skewness/kurtosis language, treat BC as heuristic, measure connectivity if using "percolation", and soften shape claims.
P2	Notation and factual consistency	Small errors weaken confidence in a technically ambitious paper.	Correct abundance ratios, rest/observed frequencies, triangle labels, ionization-fraction labels and completion-redshift definitions.

3. Major scientific and methodological issues

3.1 Normalized bispectrum: equation, implementation and geometry

Concern. The manuscript defines a normalized bispectrum that is described as dimensionless, but the printed power of k appears reversed relative to the Watkinson et al. convention. This could be a typesetting error or an implementation error. The distinction is critical.

Printed in the manuscript:

$$B_{\text{tilde}} = B / \sqrt{k_1 k_2 k_3 P(k_1) P(k_2) P(k_3)}$$

Common normalized form used in this literature:

$$b = B / \sqrt{(k_1 k_2 k_3)^{-1} P(k_1) P(k_2) P(k_3)}$$

With standard Fourier conventions, the printed expression is not dimensionless. If the code used the printed expression, the sign and exact zero crossings would remain unchanged, but amplitudes and angular shapes would be altered because k_3 varies with opening angle. Squeezed and folded configurations would be affected most strongly.

Required action. Inspect the production script or BiFFT call, record the exact normalization used, and compare it line by line with the manuscript. If only the equation is wrong, state this explicitly in the revision. If the analysis is wrong, regenerate all normalized-bispectrum figures and re-evaluate every amplitude-based interpretation.

The triangle labelling also needs correction. For the non-isosceles family $k_2 = 2 k_1$, the point $\cos(\theta) = -0.5$ is not equilateral. Figure captions should not mark it as such. Likewise, an isosceles family with variable k_3 should be written $B(k, k, k_3)$, not $B(k, k, k)$.

Interpretive change. A trough near the equilateral configuration is consistent with compact and approximately isotropic structures, but it does not uniquely confirm spherical bubbles. A zero crossing can define a useful effective Fourier scale, but it is not automatically a direct radius measurement.

- Validate the estimator using synthetic fields of identical spheres, broad radius distributions, ellipsoids, clustered bubbles, percolated regions, density-weighted fields and late neutral islands.
- Report recovered scale versus known input scale, with bias and scatter.
- Show real-space and redshift-space bispectra separately at least for representative redshifts and scales.
- State whether the quoted zero crossing is interpolated, fitted or taken from a discrete angular bin, and propagate its uncertainty into the inferred scale.

3.2 Spatial Ly-alpha background: shell weighting and energy accounting

Concern. Appendix A describes the contribution from a fixed emission redshift as proportional to the SFRD averaged inside a sphere of radius R , implemented with a spherical top-hat window. At a fixed light-travel time, however, contributing sources lie on a shell. The written method may therefore be using a cumulative volume average where a shell kernel or a difference of cumulative top-hats is required.

This is a conditional criticism: the code may already use shell differences while the manuscript describes the method imprecisely. The paper should establish which case applies. A thin-shell convolution has a $j_0(kR) = \sin(kR)/(kR)$ form, whereas a cumulative top-hat averages all sources interior to R and can repeatedly count the same interior emission as the radial integral is stepped outward.

Why it matters. An overly cumulative kernel would smooth and bias the spatial J_α field. That can change the hydrogen coupling epoch, the local 3He^+ coupling around AGN, the brightness-temperature VIDs and the higher-order statistics.

Required tests. Derive the discrete estimator in the paper and compare the FFT result with direct source summation on a small grid. Include a point-source radial profile, a uniform emissivity test, shell-number convergence and a periodic-boundary test when the Lyman horizon is comparable to half the box.

The X-ray energy budget also needs a single explicit accounting equation. The main simulation assumes $f_{\text{heat}} = 1$, while the appendix discusses Ly-alpha photons generated by secondary electrons. If the same photoelectron energy is counted both as complete heat deposition and as secondary excitation, energy is double counted. State whether secondary-excitation Ly-alpha photons are included, omitted or added only in post-processing, and ensure that the channel fractions sum to unity.

3.3 Quantitative 3He^+ amplitude and external validation

Concern. The global 3He^+ signal is weak and qualitatively reasonable, but the power-spectrum amplitude appears substantially above recent weak-coupling calculations at comparable redshift and scale. This difference may be physical, but it is currently unexplained.

A direct comparison should be made in identical units, at matched k and matched helium ionized fraction. The paper should avoid relying on visual comparison alone. Export the numerical curves and tabulate the amplitude ratio to McQuinn and Switzer-style calculations, Takeuchi et al. and the recent Spina et al. prediction.

Required signal decomposition

- Saturated-coupling reference: $x_{\text{HeII}}(1 + \delta_b)$ with the thermal prefactor set to unity.
- Density-only and ionization-only fields.
- Collisional coupling only and Ly-alpha coupling only.
- Uniform J_α versus spatially varying J_α .
- Real-space versus redshift-space signal.
- Diffuse-IGM-only versus all gas.
- Cross-spectra and correlation coefficients with density, x_{HeII} , x_{HeIII} , J_α , $T_S - T_{\text{CMB}}$ and distance to luminous AGN.

The interpretation should follow the decomposition. If the excess power is produced by rare, highly coupled halo environments, the paper should present this as a halo-proximity or dense-gas signal rather than as a generic diffuse-IGM prediction. If it persists after masking dense gas, the difference can be attributed more confidently to source demographics and the large volume.

3.4 Dense gas, star-forming cells and the effective ISM

Concern. The 3He^+ signal is reported to be concentrated near quasars, strongly star-forming regions and dense filaments. These are exactly the environments in which the Springel-Hernquist two-phase effective ISM model is active. Its effective temperature is not a literal thermodynamic temperature appropriate for atomic collisional coefficients.

This does not automatically invalidate the result, but the paper must establish whether unresolved star-forming cells control the high-temperature VID tail, high- k power or positive bispectrum. The same issue can affect hydrogen small-scale statistics, although it is more acute for helium because diffuse coupling is so weak.

- Repeat all helium statistics after excluding star-forming gas.

- Repeat after masking resolved halos or using radial cuts around halo centres.
- Show cumulative signal variance as a function of overdensity and gas phase.
- Compare the 640^3 , 1280^3 and 2560^3 rendered fields using identical masking and Fourier conventions.
- State whether cell temperatures, densities and ion fractions are volume weighted, mass weighted or otherwise reconstructed on the grid.

3.5 Helium Wouthuysen-Field coupling

The helium spin-coupling model should be benchmarked more explicitly against the specialized 3He^+ literature. The 3He^+ resonance is offset from the much more abundant 4He^+ resonance, so the spectral field at the hyperfine-pumping resonance depends on repeated 4He^+ scattering, redshifting and the detailed line profile. The appendix presently reads like a direct extension of the hydrogen Lyman-series horizon method.

- Explain how the isotope shift and 4He^+ scattering are treated.
- Benchmark x_{α}^{He} in one-zone calculations over a grid of density, temperature, ion fraction and radiation intensity.
- Show agreement or explain disagreement with McQuinn and Switzer, Takeuchi et al. and Spina et al.
- Clarify attenuation of the 40.8-54.4 eV continuum by He II, residual H I, He I and dense absorbers.

This validation is especially important because the low-redshift on-the-fly radiation solver retains only one group above the He II edge, whereas the pumping continuum lies below that edge and is reconstructed in post-processing.

3.6 Cosmic Dawn and X-ray heating

The fiducial Cosmic Dawn result should be presented as a maximal-heating reference, not as a quantitative prediction. The paper already states that $f_{\text{heat}} = 1$ over-heats the IGM and that the early SFRD, minihalo population and Pop III contribution are unresolved. Those caveats should be moved forward into the abstract, results summary and figure captions.

The $f_{\text{heat}} = 0.3$ and 0.7 curves must be described procedurally. If they are post-processing rescalings rather than self-consistent reruns, they do not consistently update ionization, recombination, electron fraction, secondary Ly-alpha production or hydrodynamic pressure. They should be called illustrative brackets. The adiabatic curve is likewise an approximate no-heating reference rather than a strict bound.

- Separate robust $z < 10$ morphology conclusions from model-dependent $z > 10$ thermal timing.
- Avoid treating the reversed ordering $z_{\text{h}} \geq z_{\alpha}$ as a prediction of the physical source model.
- If feasible, add a small-volume self-consistent energy-partition run to calibrate the post-processing brackets.

3.7 Detectability and observational comparisons

The current manuscript infers detectability from the predicted signal crossing selected idealized noise curves. This is not an end-to-end detection forecast. Thermal noise does not include foreground filtering, calibration residuals, window-function mixing, signal loss, chromatic beam effects, flagging, mode covariance or systematic floors.

RECOMMENDED WORDING Safer conclusion

Selected LUMINA modes approach or exceed idealized thermal-noise estimates for assumed integration times. Demonstrating recoverability requires instrument-specific forward modelling of the lightcone, beam, frequency response, foreground filtering and estimator transfer function.

Current observational upper limits should be plotted and discussed separately from thermal-noise forecasts. The manuscript should not describe the signal as detectable at current integration depths unless the analysis includes a realistic recovery pipeline or cites an existing pipeline applied to the same signal model.

3.8 Volume, subvolume scatter and convergence

The 500 cMpc volume is a major strength. The terminology should nevertheless be more precise. The 125 non-overlapping 100 cMpc regions share the parent box long modes, phases and global mean; they are not independent cosmological realizations. Their spread is a useful measure of subvolume or survey-volume variation, but it does not prove convergence.

- Replace "125 independent subvolumes" with "125 non-overlapping subvolumes".

- Replace "converged distribution" with "full-volume reference distribution" unless a resolution/realization test is shown.
- Use "substantially reduced finite-volume effects" rather than "largely free from finite-volume effects".
- Distinguish box-size convergence, spatial-resolution convergence, source-population convergence and absorber convergence.

A single large volume can still contain residual super-box variance and model-dependent source variance. The paper will be stronger, not weaker, if it presents LUMINA as an unusually well-sampled reference rather than claiming formal convergence that has not been demonstrated.

3.9 Hybrid low-redshift radiation treatment

Below $z = 4.75$, the calculation switches from six radiation groups to one AGN-dominated group above 54.4 eV and replaces H I/He I photoionization with a uniform UV background. This is a reasonable computational strategy, but it is not fully multifrequency, spatially coupled H+He radiation transport to $z = 3$.

- Disclose the hybrid treatment in the abstract and main conclusions, not only in the methods and final caveats.
- Show continuity of global ion fractions, temperatures and radiation fields across $z = 4.75$.
- Vary the switch redshift and group-averaged energy/cross-section.
- If possible, compare with a smaller-volume run retaining multiple He II-ionizing groups.

Claims about helium bubble sizes, pre-heating and shell sharpness should be explicitly tied to this transport approximation because spectral hardening and energy-dependent mean free paths are not resolved by one group.

3.10 RSD, gridding and Fourier conventions

The methods state that peculiar velocities are included down to particle resolution, but the map-making procedure is not reproducible from the text. Higher-order statistics and VID tails are sensitive to the exact redshift-space mapping and to grid assignment.

- State whether redshift space is implemented through a coordinate remapping or a local velocity-gradient Jacobian.
- Explain how singular or negative velocity-gradient factors are treated and whether clipping is applied.
- Specify mass-assignment scheme, smoothing kernel, deconvolution, line-of-sight choice and whether results are averaged over three axes.
- Define the Fourier transform convention, field normalization, mean subtraction, $P(k)$ and $B(k)$ units, bin widths, mode counts and covariance.
- Clarify whether all VIDs, power spectra and bispectra use coeval cubes, lightcones, real space or redshift space.

4. Conclusions that should be retained, qualified or removed

Current claim or interpretation	Assessment	Recommended framing
The 500 cMpc volume enables robust low-k statistics and better quasar sampling.	Strong and well motivated.	Retain, but use "substantially reduced finite-volume effects" rather than "converged" or "free of box-size effects".
The late-EoR hydrogen field is strongly non-Gaussian.	Scientifically sound; novelty comes from volume and joint statistics.	Emphasize the scale, cross-validation and low-k reach rather than presenting non-Gaussianity itself as new.
Helium bubbles are systematically larger than hydrogen bubbles at matched ionized fraction.	Plausible for the adopted source model, not universal.	State "in the LUMINA source and transport model" and quantify sensitivity to quasar demographics and one-group RT.
The bispectrum zero crossing directly measures a local bubble radius.	Too strong.	Call it a calibrated effective Fourier scale and report scatter against MFP-based measures.
An equilateral trough confirms spherical bubbles.	Not unique evidence.	Use "consistent with compact, approximately isotropic structures" and validate with controlled fields.

Current claim or interpretation	Assessment	Recommended framing
The power spectrum traces the largest connected structures.	Conceptually inaccurate because connectivity is phase information.	Say that it weights scales contributing most strongly to the variance.
The helium bispectrum is positive by construction.	Mathematically false for a mean-subtracted field.	Say that positivity is an empirical consequence of the modelled density, ionization and coupling correlations.
The bimodality coefficient identifies a phase transition or percolation epoch.	Unsupported by a one-point heuristic.	Treat BC as an indicator of distribution shape; measure connected components for percolation.
The 21 cm signal is detectable at current integration depths.	Not established without end-to-end recovery.	Say that selected modes approach idealized thermal-noise estimates and require instrument-specific validation.
The full 500 cMpc distribution is converged.	Not demonstrated by one realization.	Describe the full box as better sampled and less volume limited.
The calculation self-consistently follows all H and He ionization to $z = 3$.	Needs qualification below $z = 4.75$.	Describe the low-redshift treatment as a hybrid one-group He II calculation with a uniform H I/He I UV background.
The x_{HeII} field forms shells around He III bubbles.	May conflate ion-fraction shells with brightness shells.	Show stacked radial profiles and distinguish x_{HeII} , coupling and brightness morphology.

5. Statistical interpretation and topology

5.1 Skewness, kurtosis and higher-order spectra

Skewness is related to the third central moment and can be written as a configuration-integrated bispectrum. Kurtosis is a fourth-order statistic whose connected component is related to the trispectrum, with disconnected terms depending on convention. The manuscript should not describe both skewness and kurtosis as integrated versions of the bispectrum.

5.2 Bimodality coefficient

The bimodality coefficient is a heuristic descriptor of distribution shape. A value above 5/9 does not prove two physical phases, percolation or a topological phase transition. This is especially important for helium, where the reported maximum is close to the reference value. Add finite-sample correction, bootstrap errors and a dedicated modality test if bimodality is a central claim.

5.3 Extreme helium kurtosis

Kurtosis above 10^3 near the disappearance of the signal can arise when the variance approaches zero while a handful of nonzero voxels remain. The result may be numerically correct yet physically fragile. Demonstrate convergence with voxel size, masking, zero thresholds and bootstrap resampling before interpreting it as robust topology.

5.4 Percolation and connectivity

Power spectra, MFP radii and bispectrum zero crossings do not directly measure connectivity. If the word "percolation" is retained as a measured result, add the largest connected component, spanning probability, friends-of-friends statistics, Euler characteristic or Betti numbers. Otherwise use "overlap", "merging" or "late-stage network formation".

5.5 Helium shells

The manuscript should distinguish an ion-fraction shell from a brightness shell. During the main He II reionization epoch, helium outside He III regions may already be predominantly He II, so a brightness shell can arise from the combination of an He III central hole, locally enhanced coupling at intermediate radius and weak coupling far away even when x_{HeII} itself is not shell-like. Stacked profiles around quasars are the cleanest test.

6. Section-by-section revision guide

6.1 Abstract

- Retain the 500 cMpc volume, joint H/He evolution and multi-statistic analysis as the central contribution.
- Replace "robust predictions" with "large-volume reference predictions" or "predictions with reduced finite-volume effects".
- Remove the claim that the power spectrum traces connected structures.
- Describe the bispectrum scale as calibrated and effective rather than direct.
- State that Cosmic Dawn heating and quantitative 3He^+ amplitudes are model dependent.
- Mention the hybrid low-redshift one-group plus UVB treatment in one concise clause.

6.2 Introduction

- Correct the 3He abundance statement: the commonly used value is approximately 10^{-5} relative to hydrogen, not relative to 4He .
- Distinguish the 8.67 GHz rest frequency from the observed 1.4-2.2 GHz band at $z = 3-5$.
- Present 3He^+ as a difficult, exploratory tracer rather than a near-term analogue of 21 cm.
- Add the recent weak-coupling prediction as a quantitative benchmark and explain the specific novelty of LUMINA relative to it.

6.3 Methodology

- Correct and fully define the normalized bispectrum, the underlying fluctuation field and all Fourier units.
- Derive the shell-based Ly-alpha estimator and describe the code rather than only the continuous equation.
- Document X-ray energy partition and the status of secondary excitation.
- Define gas masks, star-forming cells, halo treatment, grid assignment, RSD mapping and resolution tests.
- Describe the $z = 4.75$ switch as a hybrid low-redshift approximation and justify the group-averaged quantities.

6.4 Global signal and VID results

- Call the f_{heat} variants and adiabatic curve illustrative post-processing brackets unless they are full reruns.
- Use a single explicit hydrogen-reionization completion criterion throughout the paper.
- Separate map colour-scale clipping from true VID tails.
- Add uncertainty estimates to skewness, kurtosis and BC, especially as the variance becomes small.
- Avoid calling the BC maximum a phase transition or percolation measurement.

6.5 Power-spectrum results

- Separate current upper limits from idealized forecast noise curves.
- Remove current-detectability claims without end-to-end recovery.
- Quantify grid-resolution suppression and the k range considered scientifically reliable.
- For helium, add the coupling and gas-phase decomposition and compare numerically with recent predictions.

6.6 Bispectrum results

- Correct notation and non-isosceles figure markers.
- Rebuild the physical interpretation after the normalization audit.
- Use "consistent with" rather than "confirms" for shape interpretations.
- Show estimator uncertainties and mode counts.
- Explain why a positive field does not imply a positive bispectrum and remove "positive by construction".

6.7 Bubble-size analysis

- Call MFP, power-peak and zero-crossing scales different effective statistics, not interchangeable radii.
- Calibrate each estimator on controlled fields and report bias/scatter.
- Avoid claiming that the power spectrum measures connectivity.

- Use actual connected-component statistics if percolation is central.
- State explicitly when the zero crossing changes from tracking ionized regions to residual neutral or He II islands.

6.8 Discussion and conclusions

- Organize conclusions into robust large-scale topology results and model-dependent thermal/coupling results.
- Qualify the self-consistent-to- $z = 3$ claim.
- Replace convergence claims with reduced finite-volume effects.
- Position the 3He^+ results as first large-volume exploratory predictions pending coupling and dense-gas validation.
- Bring the main limitations into the conclusion bullets rather than leaving them in a final caveat paragraph.

6.9 Appendix

- Add a derivation of the shell convolution and its discrete implementation.
- Add one-zone hydrogen and helium spin-temperature benchmarks.
- State all atomic constants, isotope abundance conventions and unit conversions consistently.
- Document the exact AGN, HMXB and hot-ISM emissivity deposition and attenuation used in post-processing.
- Provide enough numerical detail that another group could reproduce J_α and T_S from public simulation fields.

7. New validation figures and tables

New output	Purpose	Minimum content
Bispectrum implementation and synthetic-field validation	Establish that the estimator and geometric interpretation are correct.	Correct normalization; code/printed-equation comparison; spheres, radius distributions, ellipsoids, clustering and density weighting; recovered versus input scales.
Ly-alpha kernel unit-test figure	Demonstrate the spatial background algorithm.	Point source radial profile, uniform-source recovery, thin-shell versus direct summation, periodic-boundary test and shell-discretization convergence.
3He^+ signal decomposition	Identify the origin of the large power and ubiquitous positive bispectrum.	Density, ionization, collisional coupling, Ly-alpha coupling, saturated-coupling reference, real versus redshift space and cross-correlation coefficients.
Gas-phase and mask sensitivity	Separate diffuse IGM predictions from effective-ISM or halo contributions.	Star-forming cell mask, halo mask, overdensity cuts, cumulative signal versus density and $640^3/1280^3/2560^3$ grid comparison.
Stacked quasar profiles	Distinguish ion-fraction shells from brightness shells.	x_{HeI} , x_{HeII} , x_{HeIII} , density, J_α , T_S , T_K and ΔT_b as functions of quasar-centric radius and luminosity.
Low-z radiation-treatment sensitivity	Quantify the impact of the hybrid one-group model.	Switch-redshift variation, group-averaged energy/cross-section variation and, if feasible, a smaller-volume multi-group comparison.
External comparison panel	Position predictions against the literature and observations.	Common units, common ionized fractions and matched k ; LUMINA versus THESAN-1 and recent 3He^+ weak-coupling predictions; observational limits separated from idealized noise forecasts.

8. Suggested replacement language

8.1 Revised positioning paragraph for the abstract

DRAFT Suggested text

We present large-volume reference predictions for the hydrogen 21 cm and 3He+ 3.46 cm signals from the LUMINA radiation-hydrodynamic simulation. The 500 cMpc volume substantially reduces finite-volume effects and samples the rare quasar environments that shape helium reionization. We analyse the voxel intensity distribution, power spectrum, bispectrum and real-space bubble statistics. The late hydrogen-reionization results provide robust predictions of large-scale topology once the spin temperature is well above the CMB. The quantitative Cosmic Dawn and 3He+ amplitudes remain sensitive to X-ray energy partition, early source resolution, spin-coupling physics and the hybrid low-redshift radiation treatment, and are therefore interpreted as model-dependent reference predictions. Across both epochs, the combined statistics provide complementary effective measures of fluctuation amplitude, non-Gaussianity and characteristic ionization scales.

8.2 Bispectrum scale paragraph

DRAFT Suggested text

We interpret the bispectrum zero crossing as an effective Fourier scale associated with the transition between fluctuations dominated by ionized-region interiors and those dominated by the surrounding emitting medium. The mapping $R_{\text{eff}} = \pi/k_{3,t}$ is calibrated against MFP-based bubble statistics and should not be regarded as a unique geometric radius. Its accuracy depends on the bubble-size distribution, source clustering, density weighting, redshift-space distortions and the stage of reionization.

8.3 Detectability paragraph

DRAFT Suggested text

For selected redshifts and wavenumbers, the predicted 21 cm power approaches or exceeds idealized thermal-noise estimates for the assumed integrations. This comparison indicates potentially accessible modes but does not by itself establish detectability. A realistic assessment requires forward modelling of the primary beam, spectral response, foreground filtering, calibration residuals, window functions, signal loss and estimator covariance.

8.4 Volume and subvolume paragraph

DRAFT Suggested text

We use 125 non-overlapping 100 cMpc subvolumes to quantify regional scatter within the full LUMINA realization. Because these regions share the long-wavelength modes and phases of the parent box, their spread is not equivalent to an ensemble of independent cosmological realizations. The comparison nevertheless demonstrates that the 500 cMpc volume samples rare tails and quasar-rich environments more completely and substantially reduces finite-volume sensitivity relative to 100 cMpc survey volumes.

8.5 Cosmic Dawn paragraph

DRAFT Suggested text

The fiducial Cosmic Dawn history should be regarded as a maximal-heating reference. Because the simulation deposits all absorbed X-ray energy as heat and does not self-consistently partition secondary-electron energy among heating, ionization and excitation, it suppresses the canonical absorption trough. The lower-efficiency and adiabatic curves are illustrative post-processing brackets rather than fully self-consistent alternative simulations.

8.6 Helium morphology paragraph

DRAFT Suggested text

The 3He+ brightness field develops shell-like structures around luminous quasars because the central He III region produces no hyperfine emission, the surrounding He II gas can experience enhanced radiative and collisional coupling, and more distant He II remains weakly coupled to the CMB. We therefore distinguish the morphology of the brightness field from that of the underlying x_{HeII} ion fraction and test the distinction using stacked radial profiles.

9. Pre-submission checklist

- [] Confirm the exact bispectrum normalization used in code and correct Eq. (4).
- [] Regenerate bispectrum figures if the production normalization differs from the intended definition.
- [] Correct all triangle labels, captions and $B(k, k, k_3)$ notation.
- [] Validate the Ly-alpha FFT kernel against direct summation and document shell weighting.
- [] Close the X-ray energy budget and eliminate any double counting of heat and secondary excitation.
- [] Benchmark hydrogen and helium spin coupling in one-zone tests.
- [] Quantify the 3He+ power difference relative to recent weak-coupling literature.
- [] Repeat helium statistics with star-forming-gas, halo and overdensity masks.
- [] Add stacked quasar profiles for ion fractions, coupling variables and brightness.
- [] Document RSD, gridding, window functions, field normalization, units, binning and covariance.
- [] Replace current-detectability language with idealized-sensitivity language or add forward modelling.
- [] Replace "independent subvolumes" and unsupported convergence claims.
- [] Add low-redshift switch and one-group sensitivity tests.
- [] Correct skewness/kurtosis, BC and percolation interpretations.
- [] Use a single, explicit completion criterion for H I and He II reionization.
- [] Correct 3He abundance, spontaneous-rate wording, rest/observed frequency and figure-label inconsistencies.
- [] Rewrite the abstract and conclusion only after all result-changing checks are complete.
- [] Ask a collaborator not involved in the analysis to reproduce the key plots from the documented pipeline.

DECISION RULE Submission threshold

The paper is ready to submit when the P0 checks are closed, the helium signal has a transparent physical decomposition, all strong claims are aligned with the demonstrated tests, and a reader can reproduce the field construction and statistics from the methods section.

10. Selected literature checks

The following papers are the most useful technical benchmarks during revision. The goal is not to force numerical agreement, but to identify which physical or numerical assumption causes any difference.

Watkinson et al. (2017, 2019, 2021) - Normalized 21 cm bispectrum conventions, physical interpretation and BiFFT implementation.

Majumdar et al. (2018); Hutter et al. (2020) - Bispectrum sign changes, topology and scale interpretation during reionization.

McQuinn & Switzer (2009); Takeuchi et al. (2014) - 3He+ spin coupling, isotope effects and expected diffuse-signal amplitude.

Spina et al. (2026) - Recent 3He+ weak-coupling power-spectrum and detectability benchmark.

Furlanetto & Stoeve (2010); Eide et al. (2020) - Secondary-electron energy partition and the effect of maximal heating.

Iliev et al. (2014) - Finite-volume requirements for reionization statistics.

Kittiwisit et al. (2016, 2022) - Noise, imaging and recovery of one-point 21 cm statistics.

Neyer et al. (2024, 2026); Giri et al. (2018) - Real-space bubble-size estimators and interpretation of MFP distributions.

Kannan et al. (2022); Garaldi et al. (2024) - THESAN hydrogen-reionization and 21 cm comparison baseline.

Shen et al. (2020); Madau & Fragos (2017) - AGN and HMXB spectral assumptions that control helium ionization and X-ray heating.

Closing assessment

The manuscript has the ingredients of a strong and widely used paper. Its numerical scale and joint treatment of hydrogen and helium are genuine strengths. The appropriate response to the issues above is not to dilute the project, but to validate the highest-risk algorithms, separate robust results from model-dependent ones, and make the interpretation more precise. A careful major revision should substantially increase the paper's credibility and impact.

Revision sign-off tracker

Use this table to assign ownership and record the evidence that closes each major workstream before the manuscript is resubmitted to the full author team.

Workstream	Owner / target date	Status	Completion evidence
Bispectrum audit		Not started / In progress / Closed	Code check, corrected equation, regenerated plots and synthetic-field validation.
Ly-alpha and X-ray accounting		Not started / In progress / Closed	Direct-summation tests, documented shell kernel and closed energy budget.
3He+ physical decomposition		Not started / In progress / Closed	Component spectra, cross-correlations and literature comparison.
Gas masks and resolution		Not started / In progress / Closed	Star-forming/halo masks and grid-resolution comparison.
Low-z radiation sensitivity		Not started / In progress / Closed	Switch-redshift and group-averaging tests.
Observational claims		Not started / In progress / Closed	Revised language or instrument-specific recovery analysis.
Statistics and topology		Not started / In progress / Closed	Corrected BC/kurtosis language and connectivity analysis if retained.
Final rewrite and independent reproduction		Not started / In progress / Closed	Updated abstract/conclusion and one collaborator reproducing key plots.